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data . . .
 1 2 3 4 5 6

20382

$$1 - C(15, 10) = \frac{15!}{10!(15-10)!} = \frac{15!}{10!5!} = \frac{15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10!}{10! \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1!} = \frac{360 \cdot 360}{520} = 3003$$

$$2 - \frac{52!}{(52-4)!4!} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48!}{48! \cdot 4 \cdot 3 \cdot 2 \cdot 1!} = \frac{6 \cdot 497 \cdot 400}{24} = 270.725$$

$$3 - C_{m,2} = 45$$

$$4 - C_{5,3} = \frac{5!}{3!(5-3)!} = \frac{5 \cdot 4 \cdot 3!}{3! \cdot 2!} = \frac{20}{2} = 10$$

$$5 - C_{30,4} = \frac{30!}{4!(30-4)!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26!}{4! \cdot 26!} = \frac{5040}{24} = 210$$

$$6 - (2 \cdot C_{10,2}) + 1 = 2 \cdot \frac{10!}{2!(10-2)!} = \frac{10 \cdot 9 \cdot 8!}{2! \cdot 8!} = \frac{720}{2} = 360$$

$$7 - C_{52,4} \rightarrow \frac{52!}{4!(52-4)!} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48!}{4 \cdot 3 \cdot 2 \cdot 1 \cdot 48!} = 270.725$$

$$C_{48,4} \rightarrow \frac{48!}{4!(48-4)!} = \frac{48 \cdot 47 \cdot 46 \cdot 45 \cdot 44!}{4 \cdot 3 \cdot 2 \cdot 1 \cdot 44!} = 194.580$$

$$= 76.545$$

$$8 - C_{5,3} = \frac{5!}{3!} = \frac{5 \cdot 4 \cdot 3!}{3!} = \frac{20}{2} = 10$$

$$9 - C_{9,4} = \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5!}{4 \cdot 5!} = \frac{3024}{24} = 126$$

$$10 - C_{8,3} = \frac{8 \cdot 7 \cdot 6 \cdot 5!}{3! \cdot 5!} = \frac{336}{6} = 56$$

$$11 - C_{n,3} = \frac{n!}{3!(n-3)!} = 7$$

$$12 - (C_{1,3} \cdot C_{4,3}) - (C_{3,3} \cdot C_{3,3}) = 15$$

$$13 - C_{10,6} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6!}{6! \cdot 4!} = \frac{5040}{24} = 210$$

$$C_{8,4} = \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4!}{4! \cdot 4!} = \frac{1680}{24} = 70$$

$$210 - 70 = 140$$

$$14 \text{ a) } C_{15,10}$$

$$b) C_{15,5}$$

$$c) C_{15,9}$$

$$d) C_{20,10} - C_{15,10}$$

$$15 - C_{8,3} = C_{8,5}$$

$$17 - C_{8,5} = \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4!}{5! \cdot 3!} = \frac{8 \cdot 7 \cdot 6}{6} = 56$$

$$C_{5,5} = \frac{5!}{5! \cdot 0!} = 1 \quad 56 - 1 = 55$$

$$18 - C_{8,2} = \frac{8 \cdot 7 \cdot 6!}{2! \cdot 6!} = \frac{56}{2} = 28$$

$$C_{8,4} = \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4!}{4! \cdot 4!} = \frac{1680}{24} = 70$$

$$70 + 28 = 98$$

$$19 - C_{16,2} = \frac{16 \cdot 15 \cdot 14!}{2! \cdot 14!} = \frac{240}{2} = 120$$

$$N = 120 - 8 = 112$$

$$20 - C_{10,2} = \frac{10 \cdot 9 \cdot 8!}{2! \cdot 8!} = \frac{90}{2} = 45$$

$$22 - C_{50,4} = \frac{50 \cdot 49 \cdot 48 \cdot 47 \cdot 46!}{4! \cdot 46!} = 230 \cdot 300$$